

PalaySense

Rice Price & Yield Forecasting System

Actual Model Evaluation Results

Random Forest vs SARIMA vs Naive / Seasonal Naive Baselines

Coverage	Provincial: Fancy Palay, Regular Palay, Yield (MT/ha) + Municipal: 12 municipalities × 8 variety-season = 96 series
Method	80/20 chronological split • TimeSeriesSplit (5 folds price/yield, 3 folds municipal) • Training-only bias (no leakage) • Walk-forward rolling-origin
Winner	Random Forest Regression selected for all 3 provincial targets (SARIMA lost — needs 5% margin) • Municipal = RF only by design
Source File	data/forecasts/metrics.json — also visible in LGU Dashboard > Model Information

Evaluation metrics: **MAE** (Mean Absolute Error, lower is better) • **RMSE** (penalizes large errors) • **R²** (1.0 = perfect, near 0/negative = high volatility) • **MAPE** (percent error) • **Bias** (train-only CV residual, added to forecasts). Baselines: **Naive** (repeat last value) and **Seasonal Naive** (value 12 months / 4 quarters ago) on the **same hold-out window** — apples-to-apples.

Note on Price R²: Fancy R² 0.026 and Regular -0.002 look low but RMSE 3.5 PhP/kg is small vs price volatility (±3.7 PhP/kg std). Yield R² 0.956 is strong where signal is stable. See Dashboard Model Info for visual comparison.

1 Provincial Hold-Out Test (20% most recent, chronological)

Each row compares the selected model against two honest baselines evaluated on the **identical hold-out window** (ml_shared/evaluation.py:81 baseline_metrics). Lower MAE/RMSE/MAPE is better; R² near 1.0 is best. Bias is estimated from **training folds only** (evaluation.py:104 estimate_bias_cv) and added to future forecasts.

1.1 Fancy Palay Price (PhP/kg) — Horizon 6 months

Model	Status	MAE	RMSE	R ²	MAPE	Bias
Random Forest 200+50-attempt, max_depth 10+attempt	SELECTED	3.02	3.55	0.026	14.55%	0.064
SARIMA(1,1,1)×(1,1,1,12)	LOST Val RMSE > RF	—	—	—	—	—
Naive (last value)	Baseline	5.01	6.12	-1.898	26.76%	—
Seasonal Naive (12 mo)	Baseline	4.69	5.40	-1.257	22.93%	—

→ RF beats Naive by 42% (RMSE 3.55 vs 6.12) and Seasonal Naive by 34% on same window. Selection rule: SARIMA must beat RF by 5% margin (train_test_split_fancy.py:418 sarima_margin=0.95), otherwise RF retained.

1.2 Regular Palay Price (Other Variety, PhP/kg) — Horizon 6 months

Model	Status	MAE	RMSE	R ²	MAPE	Bias
Random Forest	SELECTED	3.16	3.65	-0.002	14.56%	0.659
SARIMA(1,1,1)×(1,1,1,12)	LOST	—	—	—	—	—
Naive	Baseline	4.51	5.66	-1.402	23.40%	—
Seasonal Naive (12 mo)	Baseline	4.92	5.72	-1.456	23.56%	—

→ RF keeps price level via lag features — safer on trending series (train_test_split_variety.py:385).

1.3 Yield (MT/ha, quarterly) — Horizon 4 quarters

Model	Status	MAE	RMSE	R ²	MAPE	Bias
Random Forest	SELECTED	0.077	0.102	0.956	1.75%	0.013
SARIMA	LOST direct RMSE compare	—	—	—	—	—
Naive	Baseline	0.861	0.990	-3.124	19.92%	—
Seasonal Naive (4 q)	Baseline	0.562	0.704	-1.087	12.43%	—

→ Yield is the strongest signal: R² 0.956, MAPE only 1.75%. RF beats Naive ~10×. Features: train_test_split_yield.py:29 month_sin/cos, lag1-4, roll2/4, std3, qoq/yoy. Bias-corrected metrics shown (the same predictions shipped to dashboard). Raw RF/SARIMA metrics printed in training logs (train_test_split_fancy.py:225 raw vs corrected) for audit.

2 Walk-Forward (Rolling-Origin) Validation — Honest Defense Metric

Retrains on expanding windows and forecasts the next horizon only, averaged as **mean ± std across origins** (*evaluation.py:238 walk_forward_eval*). Baselines scored on **identical windows**.

Target	Model MAE	Model RMSE	Model R ²	Model MAPE	Naive RMSE	S.Naive RMSE
Fancy 6 mo, 4 origins	2.87 ±0.17	3.33 ±0.24	-0.477 ±0.757	14.13±2.5%	5.37	5.32
Regular 6 mo, 4 origins	3.56 ±0.69	4.24 ±0.67	-0.896 ±0.829	16.70±4.1%	6.10	5.24

Target	Model MAE	Model RMSE	Model R ²	Model MAPE	Naive RMSE	S.Naive RMSE
Yield 4 q, 3 origins (MT/ha)	0.158 ±0.139	0.182 ±0.147	0.862 ±0.139	3.79±3.3%	0.603	0.864

All walk-forward RMSE beat both baselines. Price walk-forward R² negative mean (expected on small 6-point windows) — judge by RMSE/MAPE.

3 Municipal Price Forecast — 96 Models (12 municipalities × 8 variety-season)

Only **Random Forest** by design (*ml_Municipal_Forecast/train_test_split_Municipal.py:107* — *TimeSeriesSplit(3, horizon 3 mo, season 12)*). No SARIMA for municipal. Results stored under *metrics.json:242 municipal.{pilar,orani,...}.{hybridpremium_dry,...}*.

3.1 Global Average (96 series) — MAE 0.582 | RMSE 0.707 | R² 0.656 | MAPE 2.71% PhP/kg

Metric	MAE	RMSE	R ²	MAPE
Mean of 96 series	0.582	0.707	0.656	2.71%

3.2 Per Municipality (average of its 8 variety-season models)

Municipality	MAE	RMSE	R ²	vs Global
Abucay	0.587	0.710	0.688	higher
Bagac	0.590	0.713	0.634	higher
Balanga City	0.570	0.695	0.675	better
Dinalupihan	0.608	0.736	0.619	higher
Hermosa	0.606	0.731	0.655	higher
Limay	0.574	0.699	0.658	better
Mariveles	0.545	0.681	0.649	better
Morong	0.543	0.650	0.730	★ best better
Orani	0.587	0.710	0.633	higher
Orion	0.623	0.772	0.607	· worst higher
Pilar	0.574	0.694	0.643	better
Samal	0.582	0.693	0.680	better

3.3 Per Variety-Season (average across 12 municipalities)

Variety-Season	MAE	RMSE	R ²	Note
Hybrid Ordinary (Dry)	0.487	0.596	0.695	
Hybrid Ordinary (Wet)	0.457	0.545	0.770	easiest R ²
Hybrid Premium (Dry)	0.874	1.059	0.607	hardest
Hybrid Premium (Wet)	0.592	0.728	0.574	
Inbred Ordinary (Dry)	0.515	0.619	0.713	
Inbred Ordinary (Wet)	0.571	0.693	0.640	
Inbred Premium (Dry)	0.540	0.672	0.658	
Inbred Premium (Wet)	0.622	0.744	0.589	

3.4 Sample Detail — Pilar (8 series, excerpt from metrics.json:243)

Variety-Season	MAE	RMSE	R ²	MAPE	Bias	Naive RMSE
Hybrid Premium (Dry)	0.801	0.958	0.760	3.52%	0.686	2.96
Hybrid Premium (Wet)	0.710	0.842	0.336	3.12%	0.134	1.36
Hybrid Ordinary (Dry)	0.531	0.650	0.636	2.70%	0.426	1.98
Hybrid Ordinary (Wet)	0.378	0.473	0.838	1.87%	-0.185	1.60

Variety-Season	MAE	RMSE	R ²	MAPE	Bias	Naive RMSE
Inbred Premium (Dry)	0.424	0.546	0.794	1.88%	-0.096	1.78
Inbred Premium (Wet)	0.546	0.677	0.684	2.33%	0.357	1.34
Inbred Ordinary (Dry)	0.593	0.676	0.631	2.93%	-0.073	1.16
Inbred Ordinary (Wet)	0.609	0.731	0.463	3.03%	-0.029	1.47

Full 96-row breakdown available in data/forecasts/metrics.json and in the app: LGU Dashboard → Model Information → Municipality breakdown (per-municipality expanders).

4 How the Numbers Were Computed

Step	What it does	Code
Features	Price: month_num/sin/cos, quarter, lags 1-6+12, roll3/6/12, std3, price_change, yield_per_ha Yield: month_num/sin/cos, quarter, yield_lag1-4, roll2/4, std3, qoq/yoy	train_test_split_fancy.py:33 train_test_split_yield.py:29
Split	Chronological 80/20 (train on 2015+ only, 2009-2014 kept for lags)	train_test_split_fancy.py:26
CV Tuning	TimeSeriesSplit(5) for price/yield, (3) for municipal; RF tries n_estimators 250/300/350, max_depth 11/12/13; SARIMA (1,1,1)×(1,1,1,12)	train_test_split_fancy.py:87 train_test_split_Municipal.py:88
Metrics	MAE, RMSE, R ² , zero-safe MAPE	evaluation.py:25 compute_metrics
Baselines	Naive & Seasonal Naive scored on same hold-out window	evaluation.py:81 baseline_metrics
Bias	Mean residual (actual-pred) from training-only CV folds — never sees test set	evaluation.py:104 estimate_bias_cv
Walk-forward	Rolling-origin: min train 50%, n_origins 4 (price) / 3 (yield,muni), horizon 6/4/3	evaluation.py:238 walk_forward_eval

Reproduce: run the training pipeline (Scripts/run_pipeline.py) — logs print RF fold RMSE, SARIMA fold RMSE, and bias-corrected vs raw metrics. PDF auto-regenerates from the fresh metrics.json.

Prepared for thesis defense — honest, baseline-beating evaluation. Lower MAE/RMSE is better • R² near 1.0 is best • Bias near 0 is best. Price volatility is expected; judge price models by RMSE/MAPE, not R² alone.